

Chapter 4: Fractions and percentages

Drill practice and mixed revision of all 12 lessons · 119 questions

NSW syllabus outcome

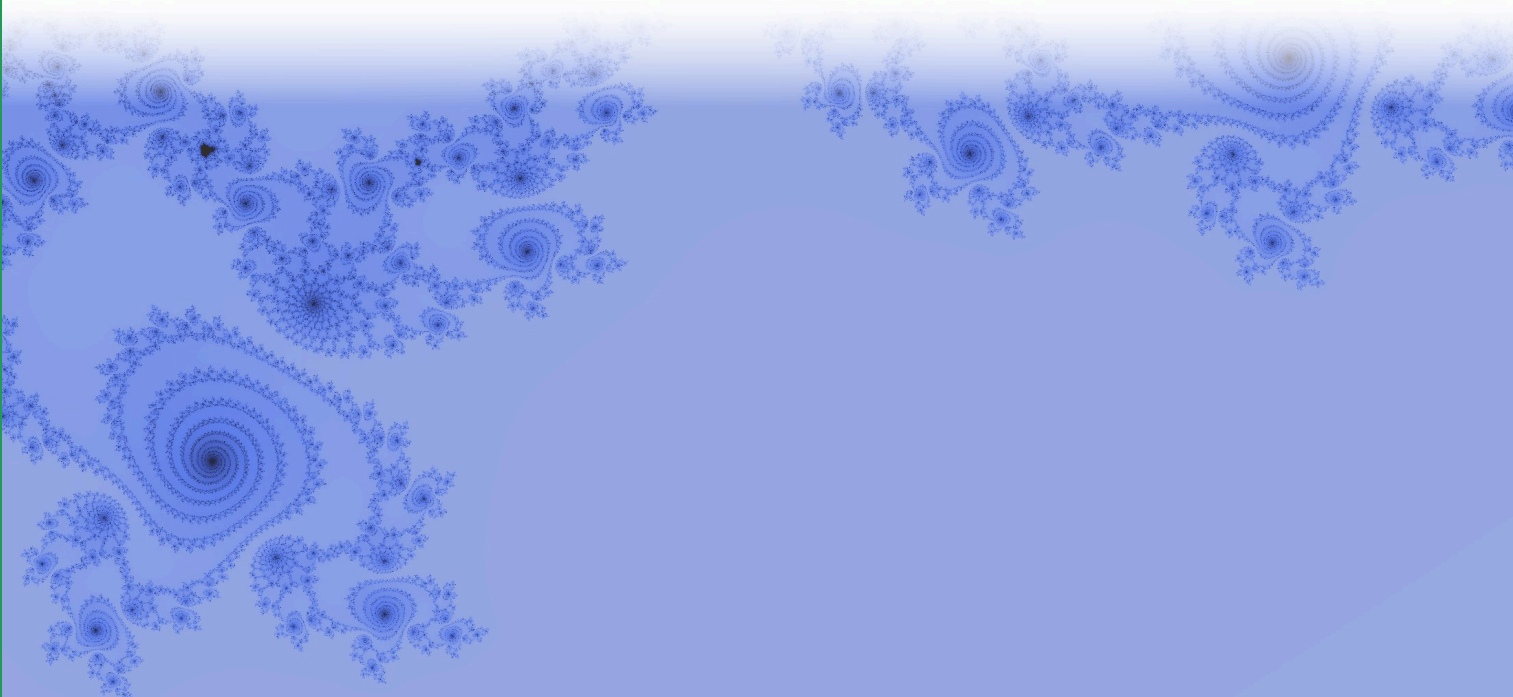
Stage 4 Number and algebra: MA4-FRC-C-01 represents and operates with fractions, decimals and percentages to solve problems.

NAME

CLASS

CHAPTER

4



Homework. Part 1 is **drill practice**: quick questions on the main skill of each lesson in Chapter 4. Write each answer on its line. Part 2 is **mixed revision**, from **Easy** to **Challenging**. Each question is tagged with its lesson, so you can look back at that lesson's summary if you get stuck.

YOU DO On your own: Part 1, drill practice.

QUESTIONS 1–38

4.01 Fractions**Easy** Simplify each fraction.

① $\frac{2}{4}$

② $\frac{2}{12}$

③ $\frac{8}{10}$

④ $\frac{16}{20}$

Medium Find the missing number in each pair of equivalent fractions.

⑤ $\frac{2}{3} = \frac{4}{?}$

⑥ $\frac{5}{6} = \frac{?}{12}$

⑦ $\frac{2}{3} = \frac{?}{9}$

⑧ $\frac{1}{2} = \frac{7}{?}$

4.02 Ordering fractions**Easy** Which fraction is larger? The denominators (or numerators) are the same.

⑨ $\frac{4}{7}$ or $\frac{3}{7}$

⑩ $\frac{4}{12}$ or $\frac{4}{6}$

⑪ $\frac{4}{12}$ or $\frac{10}{12}$

⑫ $\frac{1}{6}$ or $\frac{1}{4}$

Medium Which is larger? Use a common denominator.

⑬ $\frac{6}{8}$ or $\frac{7}{10}$

⑭ $\frac{2}{3}$ or $\frac{3}{12}$

⑮ $\frac{4}{10}$ or $\frac{1}{5}$

⑯ $\frac{1}{12}$ or $\frac{2}{3}$

4.03 Adding and subtracting fractions**Easy** Evaluate. The denominators are the same. Simplify your answer.

⑰ $\frac{8}{9} - \frac{4}{9}$

⑱ $\frac{4}{10} + \frac{8}{10}$

⑲ $\frac{4}{6} + \frac{3}{6}$

⑳ $\frac{5}{8} - \frac{1}{8}$

Medium Evaluate. Use the lowest common denominator.

㉑ $\frac{1}{5} + \frac{10}{12}$

㉒ $\frac{2}{6} + \frac{1}{2}$

㉓ $\frac{4}{6} + \frac{6}{10}$

㉔ $\frac{2}{3} + \frac{6}{10}$

4.04 Adding and subtracting mixed numerals**Easy** Evaluate. Add (or subtract) the whole numbers, then the fractions.

㉕ $5\frac{7}{9} - 2\frac{4}{9}$

㉖ $5\frac{4}{6} + 4\frac{4}{6}$

㉗ $6\frac{6}{8} - 2\frac{6}{8}$

㉘ $7\frac{3}{6} + 2\frac{1}{6}$

Medium Evaluate. Use a common denominator.

㉙ $6\frac{1}{6} + 1\frac{1}{4}$

㉚ $5\frac{2}{3} + 4\frac{2}{5}$

㉛ $5\frac{1}{2} + 4\frac{4}{5}$

4.05 Fraction of a quantity**Easy** Evaluate. Divide by the denominator.

㉜ $\frac{1}{6}$ of 30

㉝ $\frac{1}{5}$ of 25

㉞ $\frac{1}{6}$ of 48

㉟ $\frac{1}{11}$ of 110

Medium Evaluate in the unit given.

㊱ $\frac{5}{8}$ of 1 km (m)

㊲ $\frac{1}{8}$ of 1 kg (g)

㊳ $\frac{1}{2}$ of \$1 (c)

YOU DO

On your own: Part 1, drill practice.

QUESTIONS 39–74

4.06 Multiplying fractions

Easy Evaluate. Multiply the numerators and the denominators. Simplify.

39 $\frac{1}{4} \times \frac{2}{3}$

40 $\frac{6}{7} \times \frac{1}{8}$

41 $\frac{2}{3} \times \frac{5}{7}$

42 $\frac{1}{6} \times \frac{5}{7}$

Medium Evaluate. Cancel first.

43 $\frac{10}{25} \times \frac{10}{25}$

44 $\frac{6}{8} \times \frac{4}{24}$

45 $\frac{25}{35} \times \frac{15}{20}$

46 $\frac{3}{12} \times \frac{12}{18}$

4.07 Dividing fractions

Easy Write the reciprocal.

47 $\frac{10}{7}$

48 10

49 $\frac{3}{7}$

50 8

Medium Evaluate. Multiply by the reciprocal.

51 $\frac{6}{7} \div \frac{5}{8}$

52 $\frac{3}{7} \div \frac{5}{9}$

53 $\frac{2}{5} \div \frac{2}{3}$

54 $\frac{1}{9} \div \frac{4}{7}$

4.08 Fraction problems

Easy What fraction is left?

55 $\frac{4}{5}$ of a bottle of water is drunk

56 $\frac{3}{4}$ of a garden is planted

57 $\frac{3}{6}$ of a garden is planted

Medium How much is left?

58 Sam had \$50 and spent $\frac{3}{5}$ of it. How much is left?

59 Sam had \$80 and spent $\frac{1}{4}$ of it. How much is left?

4.09 Fractions on a calculator

Easy Use the fraction key on your calculator.

60 $\frac{10}{19} + \frac{1}{5}$

61 $\frac{9}{15} + \frac{3}{5}$

62 $\frac{4}{7} - \frac{3}{11}$

63 $\frac{16}{17} + \frac{3}{14}$

Medium Use your calculator. Use brackets.

64 $(\frac{1}{3} + \frac{2}{3}) \times \frac{3}{5}$

65 $(\frac{2}{6} + \frac{1}{2}) \times \frac{4}{5}$

66 $(\frac{1}{2} + \frac{5}{9}) \times \frac{1}{6}$

4.10 Percentages, fractions and decimals

Easy Write each percentage as a fraction in simplest form.

67 75%

68 25%

69 40%

70 12%

Medium Write each fraction as a percentage.

71 $\frac{18}{25}$

72 $\frac{3}{5}$

73 $\frac{9}{10}$

74 $\frac{1}{4}$

YOU DO

On your own: Part 1, drill practice.

QUESTIONS 75–89

4.11 Percentage of a quantity

Easy Find mentally.

75 50% of 152

76 10% of 310

77 25% of 12

78 10% of 420

Medium Find the percentage of the amount.

79 10% of \$360

80 25% of \$330

81 30% of \$400

82 85% of \$370

4.12 Expressing as a fraction or percentage

Easy Write as a percentage.

83 2 out of 10

84 4 out of 5

85 26 out of 50

86 25 out of 50

Medium Change to the same units, then write as a percentage.

87 20 c of 4 \$

88 10 c of 2 \$

89 6 min of 3 h

YOU DO

On your own: Part 2, mixed revision.

QUESTIONS 90–99

90

Easy

Simplify.

$$\frac{18}{24}$$
 Lesson 4.01

92

Easy

Evaluate.

$$\frac{2}{5} + \frac{1}{3}$$
 Lesson 4.03

94

Easy

Evaluate.

$$\frac{2}{5} \text{ of } 35$$
 Lesson 4.05

96

Easy

Evaluate.

$$\frac{3}{8} \div \frac{3}{4}$$
 Lesson 4.07

98

Easy

Use your calculator.

$$\frac{4}{15} + \frac{5}{6}$$
 Lesson 4.09

91

Easy

What fraction is left?

$$\frac{1}{5} \text{ and } \frac{1}{2} \text{ are eaten}$$
 Lesson 4.08

93

Easy

Complete the conversion.

$$\frac{9}{20} = ?\%$$
 Lesson 4.10

95

Easy

Write as a percentage.

$$21 \text{ out of } 30$$
 Lesson 4.12

97

Easy

Which is larger?

$$\frac{5}{8} \text{ or } \frac{2}{3}$$
 Lesson 4.02

99

Easy

Evaluate.

$$3\frac{2}{3} + 1\frac{1}{2}$$
 Lesson 4.04

YOU DO On your own: Part 2, mixed revision.

QUESTIONS 100–109

100

Easy

Find mentally.

5% of 180 Lesson 4.11

101

Easy

Evaluate.

$\frac{2}{3} \times \frac{9}{10}$ Lesson 4.06

102

Medium

Complete: $\frac{2}{7} = \frac{?}{35}$ Lesson 4.01

103

Medium

$\frac{3}{4}$ of 36 students passed. How many failed? Lesson 4.08

104

Medium

$\frac{1}{2} - \frac{1}{6}$ Lesson 4.03

105

Medium

Write 0.07 as a percentage. Lesson 4.10

106

Medium

$\frac{5}{6}$ of 72 Lesson 4.05

107

Medium

Write 21 out of 25 as a percentage. Lesson 4.12

108

Medium

$\frac{2}{3} \div \frac{1}{6}$ Lesson 4.07

109

Medium

Which is larger: $\frac{3}{4}$ or $\frac{4}{5}$? Lesson 4.02

YOU DO

On your own: Part 2, mixed revision.

QUESTIONS 110–119

110

Medium

$$3\frac{4}{5} \times \frac{5}{6}$$

Lesson 4.09

112

Medium

Find 35% of 200.

Lesson 4.11

114

Challenging

Simplify $\frac{45}{60}$.

Lesson 4.01

116

Challenging

$\frac{3}{4}$ of 2 hours (in minutes)

Lesson 4.05

118

Challenging

$$2\frac{5}{6} \div \frac{2}{3} - \frac{1}{4}$$

Lesson 4.09

111

Medium

$$6 - \frac{2}{5}$$

Lesson 4.04

113

Medium

$$\frac{5}{9} \times \frac{3}{10}$$

Lesson 4.06

115

Challenging

$$\frac{2}{3} + \frac{3}{5}$$

Lesson 4.03

117

Challenging

$$3\frac{1}{2} \div 1\frac{3}{4}$$

Lesson 4.07

119

Challenging

A \$60 shirt is 15% off. Find the sale price.

Lesson 4.11

YOU DO

On your own: extra working space. Write the question number next to your working.